A Reproducible Benchmark of Receipt Assisted Passive Feedback and Authenticated Routing in Wireless Sensor Networks

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Abstract

Routing feedback uses airtime, yet an absent observation may say little about whether a relay forwarded a packet. We examine receipt-assisted passive feedback and the transfer of frozen routing policies to an authenticated transport. A prespecified study comprises 750 IEEE 802.15.4 simulation runs on 30 new fixtures, with active-proof, naive-passive, frozen-prior and capacity-matched policy controls. Under blackholes, delivery rises from 71.28% with active proofs to 75.94% with censored passive feedback: a paired improvement of 4.67 percentage points (95% interval 2.36–6.97; Holm-adjusted p=0.0161). Energy per delivered byte falls, although total modeled radio energy increases slightly. A separate 180-run study evaluates attention, multilayer perceptron and greedy routing with serialized neighbor state, hop authentication, replay protection and signed sink receipts. Attention delivers 81.61% of packets in benign conditions and 61.11% under selective dropping; the paired comparisons do not establish an advantage over either control. Serial-interface tests reveal a receive-time freshness error, corrected before repeat testing. Compiled UART-to-radio bridge firmware runs on four emulated Cortex-M3 boards; neural inference and cryptography execute on the host PC. Each host-policy control delivers ten of ten packets. Post-review 60–300 s stress runs expose early routing-energy exhaustion and increasing service debt. The evidence supports the tested feedback replacement and verifies specific implementation paths. Physical-board operation, measured energy consumption and absolute deadline compliance remain unvalidated.

Keywords: Wireless sensor networks; Routing feedback; Authentication; Policy transfer; Reproducibility; Virtual validation

1. Introduction

Multihop sensor networks depend on link estimates, progress toward the sink and control traffic. RPL, expected transmission count and the Collection Tree Protocol use these ingredients in different ways [1, 2, 3]. Adding forwarding evidence changes the balance: an acknowledgment can improve confidence in a relay while consuming airtime that would otherwise carry data. Assessing a feedback mechanism therefore requires measuring both the information it provides and the traffic it creates.

Watchdog monitoring and explicit acknowledgments have long been used to detect forwarding misbehavior [4, 5]. Their observations need careful interpretation. Overhearing confirms a transmission attempt, while reception by the intended next hop remains uncertain. Silence is also ambiguous: the first hop may have failed, the channel may be busy, or the observer may have missed a

frame. Penalizing every silence can redirect traffic even when no node is malicious. We study how routing should handle this missing evidence; we do not train an intrusion classifier.

Recent studies combine trust with edge-assisted routing [6], optimizer-based next-hop selection [7], learned trust estimates [8], failure-aware deep Q-learning [9] and opportunistic forwarding [10]. Trust-gated reinforcement learning [11] and secure deep reinforcement learning [12] address related interactions between routing and security. Table 1 compares their stated mechanisms with the question examined here. Differences in channels, workloads and attack models prevent a defensible numerical ranking against those published results.

Table 1. Literature and experimental focus

Study	Verified mechanism	Relation to this experiment
Watchdog [4]	Overheard forwarding attempts	Established passive observation; reception ambiguity remains
2ACK [5]	Reverse acknowledgment scheme	Explicit feedback adds traffic; no numerical cross-study ranking
TARM [6]	Edge assisted trust routing	Trust-aware routing context; different state and deployment
PathGuard [7]	Trust and optimizer next-hop selection	Publisher abstract verified; not independently replicated
SMART [8]	Learned multi-feature trust estimation	Detection task differs from forwarding evidence preference
FRDR [9] and RLFS-OR [10]	Failure-aware and opportunistic learning	Different radio and duty-cycle assumptions
TG-RL-SR [11] and secure DRL [12]	Trust gated and secure routing	Recent context; claims not treated as matched experimental effects
This study	ACK gated passive evidence; four controls	Frozen 30-fixture evaluation with retained negative stages

Our earlier architecture comparisons found no joint advantage over a multilayer perceptron of similar capacity. A subsequent 600-run telemetry study reached the same conclusion after correction for multiple testing and exposed the traffic cost of active proofs. Those results, including their raw outputs, are retained. They motivated the present experiment: can receipt-assisted passive feedback replace active proofs while staying within prespecified margins against a frozen-prior control on new fixtures?

1.1 Benchmark contribution

We compare feedback rules that differ in how they admit and retain evidence. A first-hop acknowledgment makes an observation eligible for credit; each record retains its strongest observation, and absent observations remain censored. Packet-reception callbacks connect these rules to simulated radio events. The contribution lies in the controlled comparison and its traceable implementation, building on established watchdog, censoring and learning methods. Figure 1 locates the simulator-held state, and Fig. 2 shows the observer's event logic.

The benchmark tests blackhole delivery against active proofs under the prespecified multiple-testing rule. Delivery, critical-service and energy-efficiency margins against frozen-prior attention assess whether the feedback mechanism can be replaced. Naive-passive and capacity-matched controls test attribution to censoring or policy architecture. These comparisons distinguish a useful replacement from a unique algorithmic advantage.

A second implementation stage examines whether the frozen policies can use authenticated messages and node-local transit records. It comprises 180 new radio runs and separate serial-interface and firmware-emulation checks. Sections 2–4 report the feedback benchmark; Section 5 describes the authenticated transport and evaluates it separately. Its changed observation model prevents carrying the earlier performance claim into this branch.

2. System model and methods

2.1 Radio traffic and threat model

The simulated network has 100 stationary sensors in a 150 × 150 m field and a sink at (75, 75) m. The ns-3.44 low-rate wireless personal area network model provides unslotted access, MAC acknowledgments and packet reception [13]. The log-distance model uses exponent 3 and a reference loss of 40 dB, increased to 54 dB in the weak-link setting. A separate 50 m range limit applies. These settings describe the simulated radio environment.

Runs last 4 s. After a 0.4 s warm-up, the source offers 60 packets at 20 ms intervals, shortened to 2 ms for the congested burst. The three traffic classes have probabilities 0.35, 0.45 and 0.20, with deadlines of 90, 350 and 1,500 ms. Each 64-byte application frame contains 12 identifier bytes and 52 payload bytes; MAC and physical-layer overhead is additional. Nodes broadcast application telemetry every 0.5 s with a 3 ms node offset. We call these messages beacons, although they are distinct from IEEE superframe beacons. Table 2 lists the shared settings.

Table 2. Frozen network settings

Parameter	Value
Sensors and sink	100 sensors; one central sink
Field and range	150 × 150 m; 50 m hard range
Radio engine	ns-3.44; unslotted IEEE 802.15.4
Propagation	Log distance exponent 3; reference loss 40 or 54 dB
Run and offered load	4 s; 60 packets; 0.4 s warm-up
Source intervals	20 ms normal; 2 ms congested
Payload and identifiers	52 and 12 bytes; 64-byte application frame
Telemetry broadcast and cache	0.5 s period; 3 ms/node staggering; 0.75 s expiry
Feedback expiry and poll	150 ms; 1 ms
Traffic classes	Probabilities 0.35, 0.45, 0.20; deadlines 90, 350, 1,500 ms
Blackhole ratio	0 or 25%; non-forging forwarding faults
Assumed voltage and currents	3 V; TX 17.4 mA; RX/listen 18.8 mA; off 20 µA
Policy parameters	Attention 63,368; MLP 63,554; fixed weights
Holdout	30 fixtures × 5 settings × 5 arms = 750 runs

For each blackhole fixture, 25% of sensor identities are selected independently as faulty relays. They accept packets but skip the forwarding decision, while retaining the modeled beacon and receipt behavior. The threat model excludes forged control values, collusion and a compromised sink. Attack labels are used only by the simulator and evaluator; the router sees forwarding observations, not those labels.

2.2 State construction and frozen routing policy

A received beacon creates or refreshes an observer-specific cache entry containing energy fraction, pending-request pressure, timestamp and link-quality indication. Entries expire after 0.75 s. The current node, visited nodes, exhausted nodes and nodes beyond 50 m are excluded from relay selection. The following fixed score retains at most eight candidates:

$$S_{uv} = 1.25\, g_{uv} + 0.75\, q_{uv} + 0.25\, e_v - 0.55\, R_{uv} - 0.20\, Q_v \tag{1}$$

• g is sinkward progress divided by 50 m; q is estimated MAC success probability; both are dimensionless.

• e is remaining modeled energy divided by 0.5 J, Q is pending-request pressure normalized by 32, and R is the evidence-risk upper proxy; all are dimensionless.

The observation contains 13 context values and eight candidate rows of ten values each. Context records local energy and queue state, normalized sink distance, remaining deadline, traffic class and six source-service debts. A candidate row records whether the node is the sink, its energy fraction, estimated MAC success, received link quality, queue pressure, sinkward progress, evidence risk and uncertainty, expected retries and normalized link delay. During radio transfer, received link-quality indication replaces the RSSI proxy used in training. The weights remain fixed despite this feature shift.

The coefficients in Eq. (1) are fixed hand-set preselection constants, rather than learned parameters. The retained source records their values but supplies no systematic tuning record. They are treated as benchmark settings. Equation (2) divides the query–key product by the square root of 96 in both the native equation and the PyTorch implementation; multiplication by that factor is not implemented.

Shared encoders map context x and candidate h to separate 96-dimensional representations. Their query–key interaction and a shared candidate bias give the action logit:

$$\ell_{uv} = \frac{q_\theta(x_u)^T k_\theta(h_{uv})}{\sqrt{96}} + b_\theta(huv) \tag{2}$$

• ℓ is a dimensionless action logit; x and h are the context and candidate feature vectors.

• qθ and kθ are learned query/key transforms and bθ is a learned candidate bias; θ is fixed throughout transfer.

A masked argmax chooses the relay. Candidate encoders share their weights, and the critic pools valid candidates by a masked mean, following the set-function motivation in [14]. The attention model has 63,368 parameters; the capacity-matched perceptron has 63,554 parameters and 203 hidden units. Candidates are ranked by the preselection score, with configured node identity breaking ties. Shared scoring therefore does not make the entire routing pipeline invariant to arbitrary node relabeling.

Both model families were trained on seeds 41001–41030 for 400 episodes per seed. Training used eight updates of 50 episodes, each followed by three full-batch PPO epochs. The learning rate was 0.0003, clipping parameter 0.2, discount 0.99, generalized-advantage factor 0.95 and entropy coefficient 0.01 [15, 16]. Six cost channels represented class-specific delay violations and losses. Lagrange multipliers used an update rate of 0.03 and were clipped to [0, 50]. This finite-budget procedure does not inherit the guarantee of constrained policy optimization [17]. Checkpoints and thresholds were held fixed throughout the new radio experiments.

2.3 Receipt assisted censored evidence

Observer u keeps one record for selected relay v and packet p. It stores the expected first-link success, ACK status, expiry time, strongest observation and credit already assigned. A matched overheard data frame has strength 0.5; a reverse receipt generated by sink delivery has strength 1. Repeated receptions cannot accumulate credit because the record retains only the maximum strength:

$$s_{uvp} \leftarrow \ max\{s_{uvp},\ 0.5\, I_{overheard}, I_{receipt}\} \tag{3}$$

• s is the strongest evidence strength, dimensionless and bounded by 1.

• I_overheard and I_receipt indicate matched received forwarding frames and reverse receipts, respectively; they are dimensionless binary observations.

An observation earns credit only after the observer receives a successful MAC acknowledgment for its own transmission, and only for strength not credited previously. Evidence that arrives before the acknowledgment is stored until the record becomes eligible. The positive update is:

$$\alpha_{uv} \leftarrow \alpha_{uv} + \ clip(q_{uv} max\{0, s_{uvp} - c_{uvp}\},\ 0.05,\ 1), c_{uvp} \leftarrow s_{uvp} \tag{4}$$

• α is a dimensionless effective success count, initialized to 1; c is the credited record strength, initialized to 0.

• q is the dimensionless link-success estimate at registration. The clipping floor applies to each positive increment, not to their sum.

In the censored arm, the effective failure count β remains at its prior of 1. A failed first hop closes the record without positive credit. If the first hop is acknowledged but no forwarding evidence arrives within 150 ms, the record expires without a failure increment; a later receipt can still upgrade it. For $q<0.1$, the clipping floor can make two half-credit updates exceed a single full q-weighted update. The smallest q in the audited decisions was 0.6923, so this case did not occur. The effective counts should nevertheless not be interpreted as a calibrated posterior.

Equation (4) applies clipping only when the uncredited evidence increment is positive. Its static floor is a heuristic and can alter the total assigned credit when one observation is split into successive increments. It is not a Bayesian count update. In Eq. (5), 1.645 is likewise a fixed score multiplier, not a calibrated 95% coverage guarantee or a Beta quantile. Replacing either rule changes the feedback mechanism and requires a separate experiment.

The policy retains its existing evidence-risk coordinates. With the rate-anomaly term disabled, the mean and upper proxy are:

$$\rho_{uv} = \ 0.75\left(1\ - \frac{\alpha_{uv}}{\alpha_{uv}+\beta_{uv}}\right), R_{uv} = \ clip(\rho_{uv} + \ 1.645\ \times\ 0.75\sigma_{uv},\ 0,\ 1) \tag{5}$$

• ρ and R are dimensionless risk coordinates.

• σ is the beta-form standard deviation $\mathrm{sqrt}[\alpha\beta / ((\alpha+\beta)^2(\alpha+\beta+1))]$, used as a dimensionless uncertainty proxy rather than a calibrated maliciousness interval.

Table 3 sets out the feedback controls. Naive passive uses the same observations and duplicate suppression but penalizes an acknowledged forwarding attempt when evidence is still missing at expiry. Active proofs use separate packets and the earlier timeout convention; their credit rule does not require the passive arm's first-hop ACK gate. Frozen prior keeps the initial risk coordinates without updating them. The capacity-matched MLP uses the censored ledger. Thus, the active-versus-censored comparison changes feedback semantics as well as traffic; it does not isolate airtime alone.

Table 3. Feedback controls

Arm	Evidence and expiry	Policy and interpretation
Active proofs	Separate proofs; legacy timeout and ACK eligibility	Attention; joint feedback and traffic comparator
Naive passive	Overhear/receipt positives; missing ACKed forwarding penalized	Attention; missing-observation control
Frozen prior	Same initial risk coordinates; no evidence updates	Attention; score-update control
Capacity MLP	Censored ledger with same radio observations	63554 parameters; representation control
Censored passive	Own ACK required; max-strength credit; missing evidence censored	63368-parameter attention; proposed replacement

2.4 Source service state and simulator boundaries

For each class, the source updates a service debt using the virtual-queue form associated with long-run constraint tracking [18]:

$$Z_c \leftarrow \max\{0, Z_c + I_{violation} - b_c\} \tag{6}$$

• Zc is a dimensionless source-service debt and bc is the allowed violation fraction for class c.

• I_violation is 1 for a missing or late delivered packet and 0 otherwise; loss has a separate analogous channel.

In this benchmark, only the source updates service debts, after receipt or a 1.5 s expiry. Intermediate relays do not maintain a complete debt for transit service. A shared visited set prevents loops, and the simulator keeps the reverse-path map used for receipts. Neither mechanism serializes its state dissemination. Beacon energy and queue values also come from metadata keyed to the transmitted identifier: a radio receive callback controls access, but the values themselves are not decoded from the frame. Figure 1 separates these simulator inputs from the radio event path. The authenticated branch in Section 5 replaces this state interface.

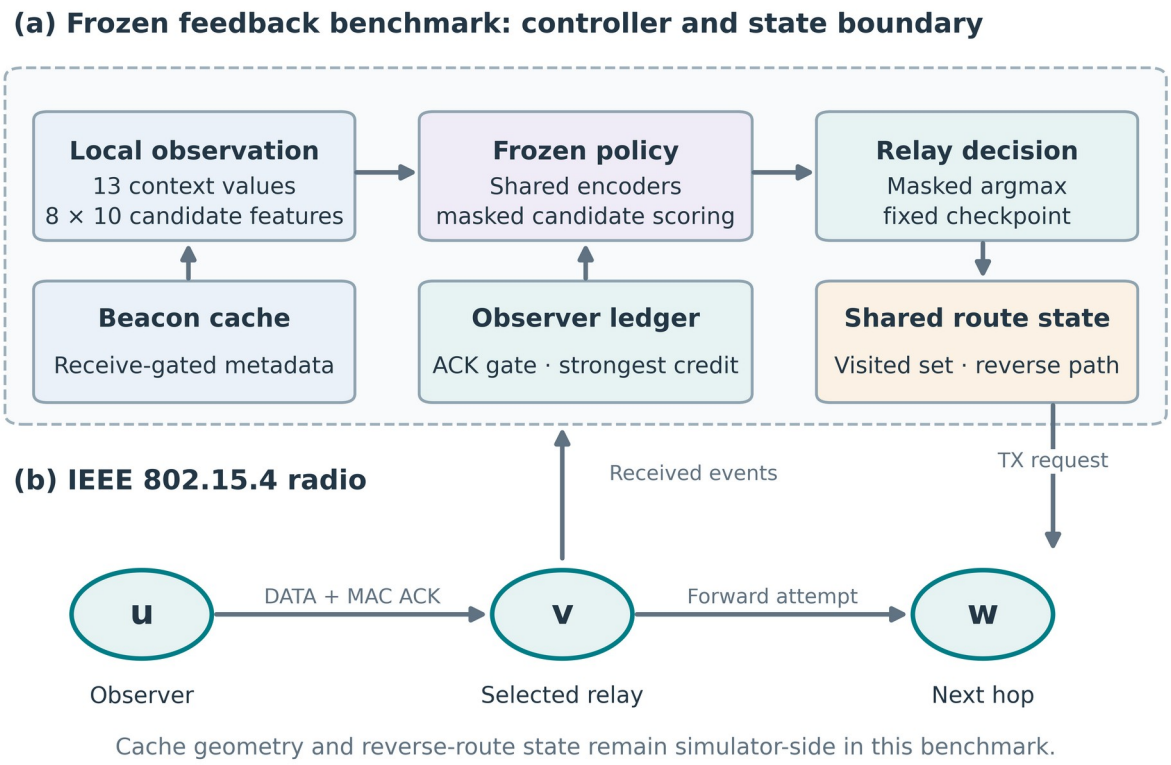


Fig. 1. Frozen benchmark architecture.

Algorithm 1 gives the observer procedure, with the event flow shown in Fig. 2. Radio callback controls check reception of a two-hop forwarding observation, absence of reception at an out-of-range observer and absence of proof frames in the passive arm. They verify those callback paths, rather than authentication or intrusion-classification accuracy.

Algorithm 1: Observer feedback and routing

Input: received MAC events, cached telemetry, fixed policy θ, ledger mode and 150 ms timeout.

Output: next-hop requests and updated observer evidence coordinates.

Steps:

1. At a routing event, reject a visited or exhausted node and enforce the 22-hop cap.
2. Construct receive-gated candidates, retain at most eight and evaluate the masked fixed policy.
3. Register the observer–relay–packet record with expected link success; send the data frame.
4. On own first-hop ACK success, mark the record eligible; credit any stored uncredited observation.
5. On ACK failure, close the passive record without success or failure credit.
6. On a matched received forward attempt, retain maximum strength 0.5.
7. On a matched received reverse delivery receipt, retain maximum strength 1.0.
8. Apply a positive update only if eligible and strength exceeds credited strength; deduplicate repeats.
9. At each 1 ms poll, close expired records; censor missing evidence in the proposed arm.
10. Allow a later receipt to upgrade an acknowledged censored record; keep failure count at its prior.
11. Update source service debt on receipt or service expiry; retain the frozen policy for later decisions.

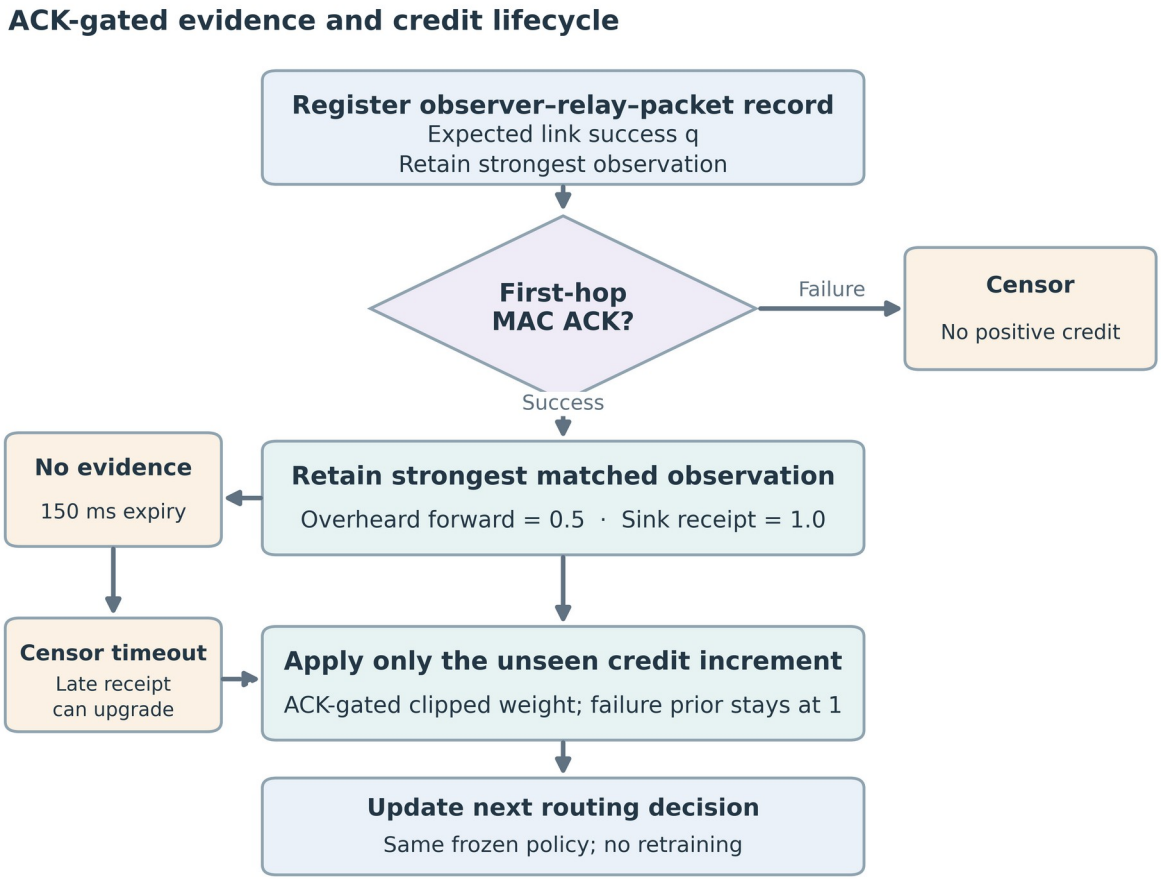


Fig. 2. ACK-gated evidence lifecycle.

3. Experimental design

3.1 Development separation and untouched fixtures

Figure 3 traces the experimental stages. The experimental protocol pipeline records source freezing, the technical pilot and all retained runs. Telemetry fixtures 76001–76030 informed the redesign. The protocol, source and decision rule were committed as 4027f071322cb5682f39329f656d76df3b68ff62 before evaluating fixtures 77001–77030. Technical pilot 76901 is excluded from inference. Each new fixture uses the unchanged attention and MLP checkpoints from seed 41001 plus its fixture offset.

The four-second horizon isolates a burst and its feedback. Positions are available at startup and telemetry begins immediately; the model does not simulate RPL or CTP parent discovery. This design therefore evaluates a transient workload rather than routing convergence or steady operation.

Thirty fixtures crossed with five settings and five feedback/policy arms yield 750 runs and 45,000 offered source packets. No run is excluded. Within a fixture, arms share topology, traffic classes, offered packets and blackhole identities. Their event schedules can diverge, so matching seeds does not guarantee identical radio draws at every event. Statistical pairing therefore applies to the complete fixture and checkpoint. Packets are not independent replicates, and the sampled geometries come from one stationary placement family with a central sink.

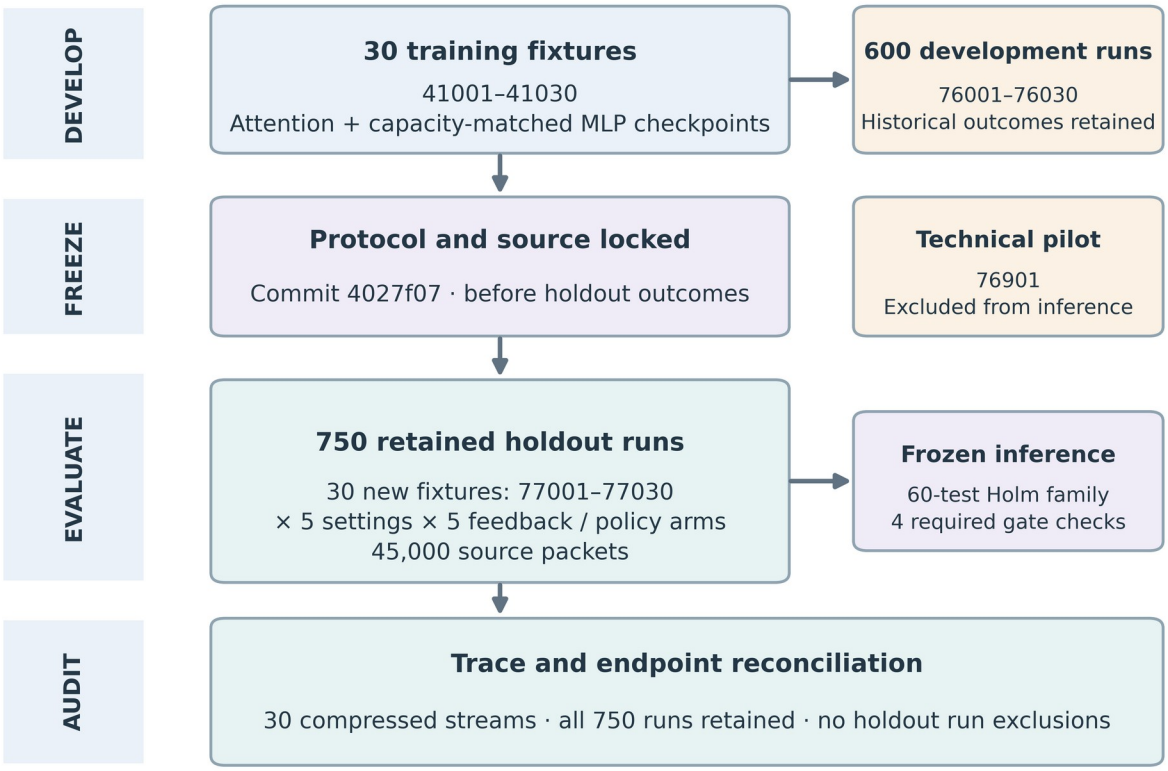


Fig. 3. Frozen evaluation stages.

3.2 Endpoints and radio accounting

Delivery is the fraction of offered source packets received by the sink:

$$D = \frac{N_{delivered}}{60} \tag{7}$$

• D is dimensionless; Ndelivered counts distinct packet identifiers received by the sink during the 4 s horizon.

For critical class 0, a packet violates its requirement if it is lost or arrives after 90 ms:

$$V = \frac{\sum_{p \in C_0} I[p \ missing \ \vee delay_p > 0.09 \ s]}{|C_0|} \tag{8}$$

• V is a dimensionless violation fraction and C0 is the set of offered critical-class packets.

• delayp is sink reception time minus source generation time, in seconds.

PHY-state integration sums the time each sensor spends transmitting, receiving/listening and switched off, including control traffic and retries. Sink energy is recorded separately and omitted from the sensor numerator. With the assumed 3 V supply, modeled radio energy is:

$$E = 3 \sum_{u \in sensors} \left(0.0174\, T_{u,TX} + 0.0188\, T_{u,RX/listen} + 0.000020\, T_{u,off}\right) \quad (9)$$

• E is modeled sensor radio energy in joules; residence times T are in seconds and coefficients are amperes.

• The supply is 3 V. Currents are explicit assumptions, not calibrated measurements; CPU, sensing and cryptography are excluded.

Dividing this expenditure over the common observation window by delivered application payload gives energy per delivered byte:

$$\eta = \frac{E}{52\, N_{delivered}} \quad (10)$$

• η is radio joules per delivered payload byte; 52 is payload bytes per packet.

• A lower η can result from a larger delivered-byte denominator even if total E does not decrease.

3.3 Statistical analysis and frozen gate

For each comparator and endpoint, we calculate one difference per fixture and summarize the 30 paired differences. The two-sided 95% Student t intervals are descriptive and unadjusted. Hypothesis tests retain the prespecified Holm family of 60 comparisons: five settings, four comparators and three endpoints [19]. Delivery and violation differences use percentage points; energy-efficiency differences use J/byte. The comparison family is unchanged after examining the results.

The replacement passes only if all four prespecified checks succeed. Against active proofs, blackhole delivery must improve with Holm $p<0.05$. Against frozen prior, the one-sided 95% bounds must remain above −1 delivery point, below +1 critical-violation point and below an energy-efficiency geometric ratio of 1.02. Table 5 gives the criteria and outcomes. These relative margins assess a feedback replacement; the original absolute class-0 violation budget remains 1.5%.

4. Results

4.1 Delivery and matched controls

Under blackholes, censored passive attention delivered 75.94% of offered packets, versus 71.28% for active proofs. The paired improvement was 4.67 percentage points (95% CI [2.36, 6.97]; Holm $p=0.0161$). Figure 4 compares effects across settings, and Fig. 5 shows every paired blackhole fixture alongside the control contrasts. Only the blackhole delivery comparison against active proofs survived the 60-test correction. The improvement should therefore be interpreted for that setting.

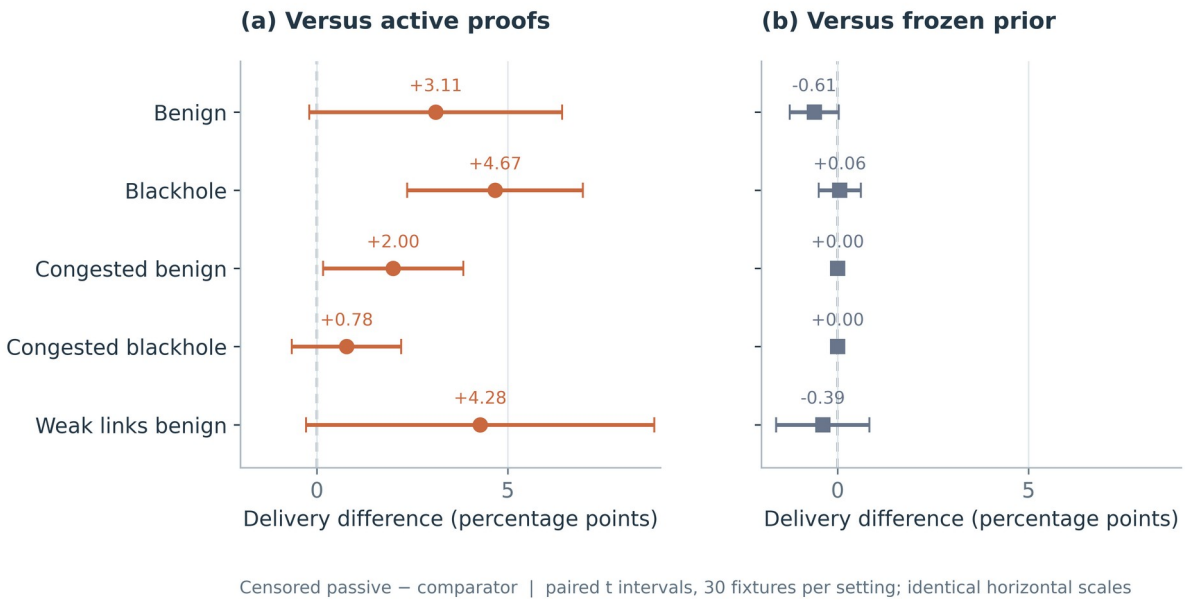


Fig. 4. Delivery effects across settings.

Table 4 reports the remaining blackhole contrasts. Delivery differed by −0.17 points against naive passive, +0.06 points against frozen prior and +0.44 points against the capacity-matched MLP. None was significant after correction. The active-proof comparison consequently does not establish a unique benefit of censoring, evidence updating or attention. The declared margins also do not establish equivalence outside the tested conditions.

Table 4. Blackhole paired effects

Comparator	Delivery points	Critical violation points	Energy mJ per byte
Active proofs	+4.667 [+2.365, +6.969] p=0.01612	-4.518 [-8.253, -0.783] p=1	-0.644 [-0.974, -0.315] p=0.02385
Naive passive	-0.167 [-1.557, +1.223] p=1	-0.017 [-2.381, +2.348] p=1	-0.066 [-0.416, +0.284] p=1
Frozen prior	+0.056 [-0.498, +0.609] p=1	-0.124 [-1.388, +1.139] p=1	-0.011 [-0.078, +0.057] p=1
Capacity MLP	+0.444 [-1.461, +2.350] p=1	-2.384 [-6.065, +1.298] p=1	-0.076 [-0.364, +0.212] p=1

Values are censored passive minus comparator; intervals are unadjusted paired 95% intervals. p values are Holm adjusted over the original 60 tests. Negative critical-violation and energy-per-byte differences favor censored passive. These are differences, not absolute arm means.

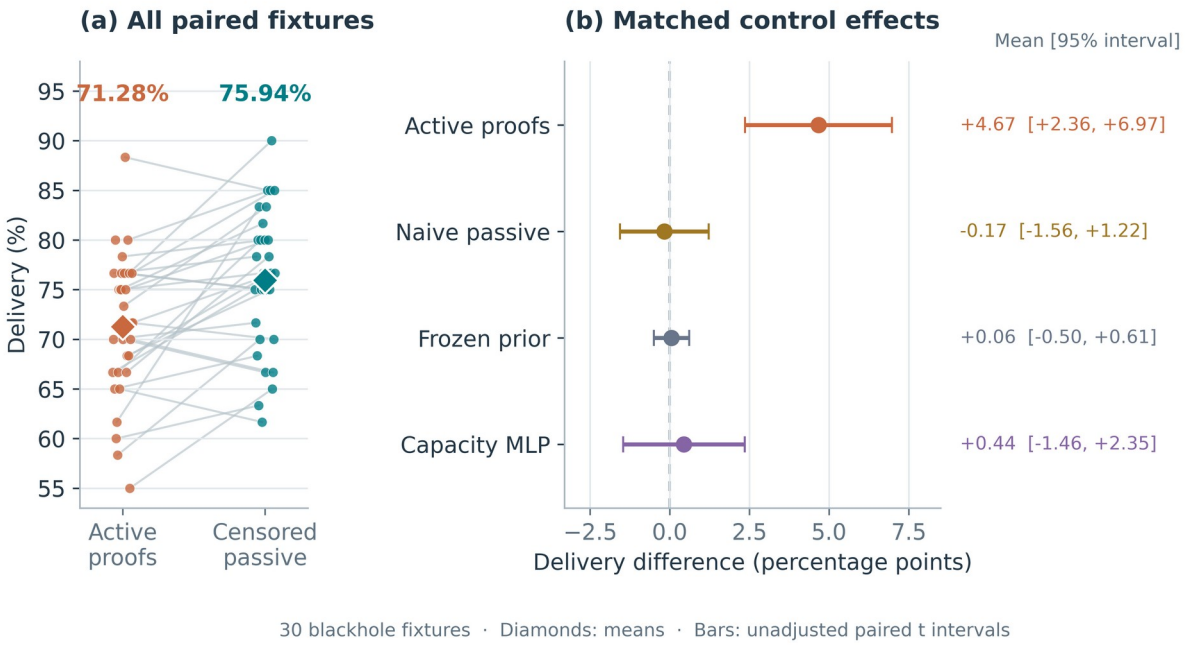


Fig. 5. Blackhole observations and effects.

4.2 Energy efficiency and control traffic

Under blackholes, energy per delivered payload byte fell by 0.0006443 J/byte relative to active proofs (95% CI [−0.0009740, −0.0003145]; Holm p=0.0239). Total modeled sensor radio energy, however, rose from 22.546943 to 22.547473 J, a descriptive increase of 0.000531 J. The gain is in delivery efficiency, not total energy saving. Across the 750 runs, receiving/listening accounts for an average 99.36% of modeled sensor radio energy.

Duty cycling changes the cost of passive observation. As an accounting example, an extra 150 ms listening window instead of radio-off time costs 3 × (0.0188 − 0.000020) × 0.15 = 0.008451 J under the assumed currents. Overlapping windows and existing awake periods reduce the incremental cost. This calculation is not a duty-cycled MAC experiment. Such a comparison must charge channel checks, rendezvous, observation windows and proof retransmissions to all arms; the always-listening results do not predict which mechanism would use less energy.

Figure 6 separates total energy, energy per delivered byte and PHY transmission count. Censored passive generates an average 60.57 fewer PHY transmissions per blackhole run than active proofs. All passive and frozen-prior arms request zero forwarding proofs, compared with 3,369 requests across the active arm's 150 runs. Beacons and reverse receipts remain in every arm. In the assumed always-listening model, receive/listen current exceeds transmit current; reducing transmissions therefore need not reduce energy over a fixed window. More delivered bytes explain the lower energy-per-byte ratio.

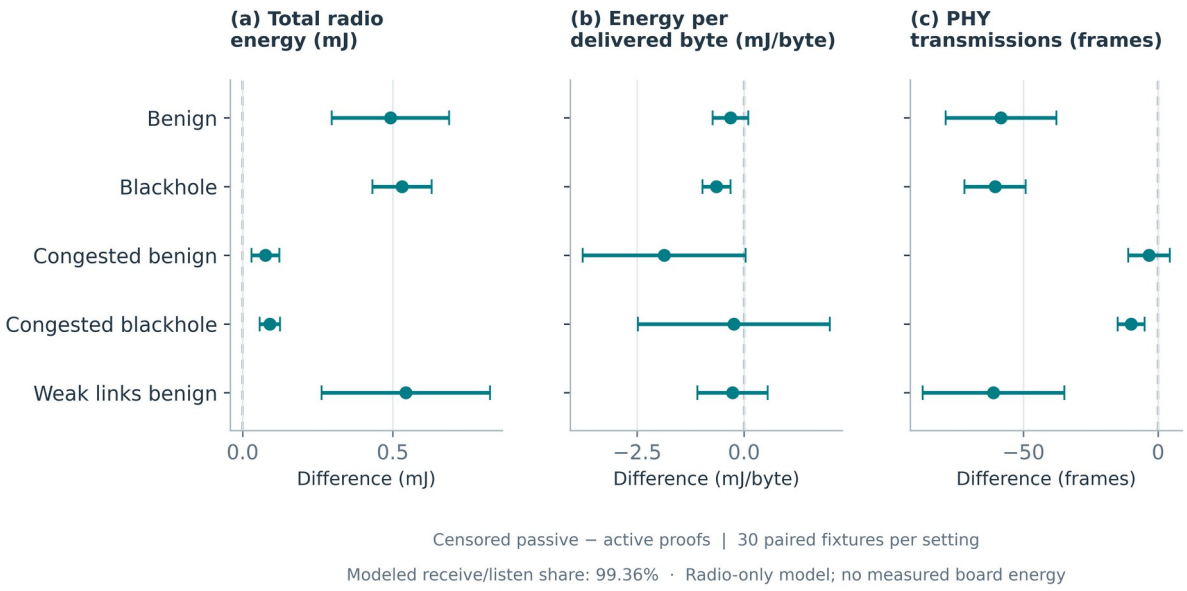


Fig. 6. Modeled radio-energy attribution.

4.3 Gate outcome and service limits

All three blackhole margins against frozen prior passed. The one-sided delivery lower bound was −0.405 percentage points, the critical-violation upper bound was +0.925 points, and the geometric energy-efficiency ratio upper bound was 1.00505. With the corrected delivery comparison against active proofs, these results satisfy all four checks in Table 5. The critical-violation bound lies close to its +1-point margin, leaving little room beyond the prespecified threshold.

Table 5. Prespecified incremental gate

Check	Frozen criterion	Observed result
Delivery versus active proofs	Positive gain and Holm p<0.05	+4.667 points; p=0.0161; pass
Delivery versus frozen prior	One-sided lower bound >−1 point	−0.405 points; pass
Critical violations versus frozen prior	One-sided upper bound <+1 point	+0.925 points; pass
Energy efficiency versus frozen prior	One-sided geometric upper ratio <1.02	1.00505; pass
Joint incremental decision	All four checks required	Pass; absolute QoS compliance not established

The reduction in critical violations against active proofs did not survive correction for multiple testing. Censored passive still violates approximately 26.2% of critical requirements under blackholes, well above the original 1.5% budget. Passing the relative replacement test therefore does not establish absolute deadline compliance or a guaranteed quality-of-service controller.

4.4 Implementation and trace audit

The action and ledger audit checked 67,522 decisions, 24,747 passive observations, 28,353 evidence updates and 75,750 node-energy identities. A separate endpoint audit reconstructed delivery and critical-violation values for all 750 runs from source decision times and sink receptions, then checked beacon counts, decision counts and PHY-state energy. All 30 gzip streams were read to completion. The original-stage suite had 51 passing tests; the expanded implementation now has 83 passing tests, including the radio callback checks.

Release checks found a truncated trace for fixture 77004. It was regenerated with the same frozen implementation, and all 750 canonical summary rows matched the saved pre-recovery summary exactly. Parameters, checkpoints and analysis rules were unchanged. The recovery record and audit code accompany the raw traces. These checks reconcile the implementation's own records; they do not provide an independent radio model, external packet capture or replay of every beacon-reception and service-state transition.

5. Authenticated transport and virtual validation

5.1 Local state and authenticated evidence

The authenticated branch uses the message path in Fig. 7. Coordinates, modeled remaining radio energy, queue state and transmit time are serialized in 58-byte application telemetry frames. A node relies on its provisioned own and sink coordinates, authenticated received cache, local MAC outcomes and packet history. Pairwise HMAC-SHA256 protects each hop with a 128-bit tag [20]. Provisioned 64-bit boot sessions and a 64-entry replay window reject stale sessions and duplicates. A source authenticator binds the immutable data fields. Ed25519 signs the source, session and packet identity at the sink [21], and the receipt follows node-local reverse records. Application frames are at most 114 bytes.

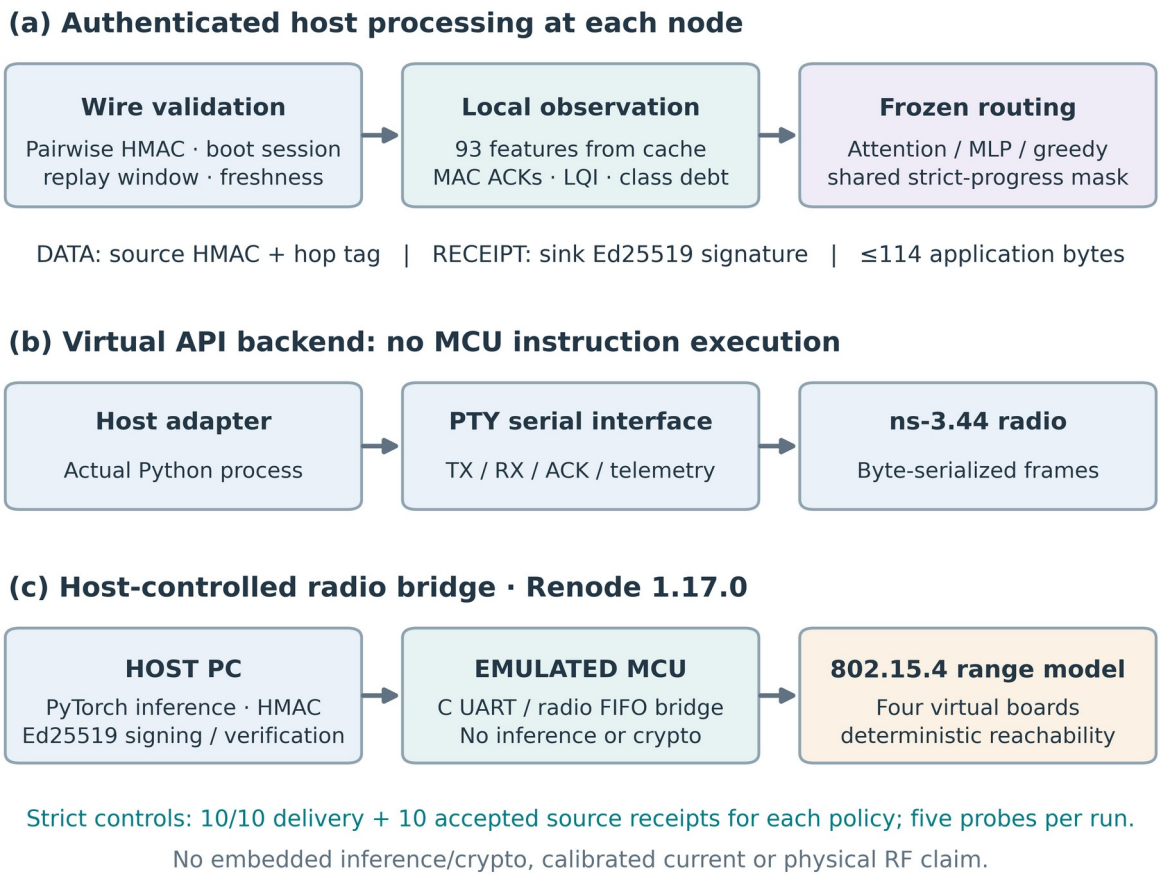


Fig. 7. Authenticated state and virtual interfaces.

Pairwise keys cannot authenticate a transmission between two other nodes, so this branch awards no credit for third-party overhearing. Positive forwarding credit requires both a successful local MAC acknowledgment and a verified sink receipt; missing receipts remain censored. Each node retains at most 128 records, with a 2 s expiry. Keys and sessions are provisioned out of band. Protected key

storage, online key exchange and verification of telemetry truthfulness after key compromise remain outside the implementation.

5.2 Frozen host policy transport protocol

The authenticated-transfer protocol and source were committed as 5f06b9213e73206aad11ec8f3a96079a17011a20 before its outcomes were observed. Seeds 83001–83030 define 30 independent topology fixtures. Each pairs a frozen attention checkpoint with its capacity-matched MLP and a greedy progress control. The six-node diamond has two first-hop choices and two downstream relays. Nominal positions are (0,0), (25,−15), (25,15), (50,−15), (50,15) and (75,0) m, independently jittered by ±2 m; range is 38 m. Benign and selective-dropping conditions give 180 runs. In the latter, relay 1 discards even-identifier data after MAC reception. The router cannot access this attack label.

Each run lasts 7.5 s and offers 60 packets carrying 16 payload bytes. Traffic starts at 0.5 s, uses 75 ms intervals and cycles through the three deadline classes. Authenticated telemetry is attempted to every provisioned peer every 0.5 s, staggered by 2 ms. The record includes out-of-range requests and their modeled radio costs. All arms enforce strict progress toward the sink and retain at most eight candidates. Greedy picks the shortest remaining sink distance; the learned policies retain their trained weights. Fixed ns-3 seeds and streams pair the fixtures, although different decisions can consume different random sequences.

The observation has 93 elements: 13 context values and eight ten-value candidate rows. Decoded telemetry supplies neighbor geometry and energy. MAC success starts from a Beta(9,1) prior, and an exponentially weighted ACK-delay estimate uses coefficient 0.2 with an initial value of 5 ms. Received LQI replaces the training RSSI proxy. Queue features use raw MAC pending-request counts instead of normalized training pressure. Forwarding risk uses a positive-only transform of signed-receipt success counts and current link reliability. Nodes maintain separate class-delay and loss debts. These substitutions change the observation distribution; the transfer neither recreates the training model nor calibrates intrusion probabilities.

5.3 Transfer results and audit

Table 6 and Fig. 8 summarize all 180 runs. In benign conditions, attention delivers 81.61% of packets, versus 82.72% for greedy and 83.72% for MLP. Its paired differences are −1.11 points against greedy (95% interval −3.39 to 0.83) and −2.11 points against MLP (−4.50 to 0.28). Under selective dropping, delivery is 61.11% for attention, 62.89% for greedy and 60.50% for MLP. The attention differences are −1.78 points (−6.50 to 2.67) and +0.61 points (−2.56 to 3.78), respectively. Intervals use 10,000 paired percentile-bootstrap samples across fixtures with seed 94001. They are exploratory, without multiplicity-adjusted superiority testing.

Table 6. Authenticated policy transfer.

Condition / policy	Delivery (%)	Receipts (%)	Critical failure (%)	Radio energy (J)
Benign / greedy	82.72	79.22	31.17	2.5188
Benign / mlp	83.72	79.72	30.83	2.5189
Benign / attention	81.61	77.61	31.17	2.5189
Dropping / greedy	62.89	60.56	46.67	2.5204
Dropping / mlp	60.50	58.22	48.50	2.5206
Dropping / attention	61.11	58.17	47.83	2.5205

Means over 30 fixtures per condition and policy. Receipt rates count source-verified sink receipts. Critical failure includes undelivered critical packets. Energy covers modeled radio residence, not board or host CPU energy.



Fig. 8. Frozen policy transfer and paired effects.

The −1.78-point attention–greedy difference under selective dropping has an interval that crosses zero, so it does not establish greedy superiority. LQI replacing the training RSSI proxy, raw queue counts and receipt-only forwarding evidence are plausible sources of observation shift; their causal contributions were not isolated. Greedy is the economical starting point for deployment assessment because it avoids neural weights and tensor inference. It still requires neighbor state, authentication and packet processing.

The transfer runs show that the frozen policies execute with serialized simulated-radio frames and multiple relay choices. They do not establish an attention advantage. Counting undelivered packets as failures, critical-deadline failure is 31.17% for benign attention runs and 47.83% under selective dropping. Total modeled radio energy is approximately 2.519 and 2.520 J per six-node run. Always-on listening and repeated authenticated telemetry dominate the radio costs, limiting differences between policies. Embedded CPU and cryptographic energy are excluded.

The audit reconstructed 27,240 decisions, including 14,001 with more than one candidate. It matched 88,538 received byte frames to earlier transmissions, verified 46,204 public sink-signature occurrences and reconciled 1,080 radio-energy identities. Candidate cache values were checked against received beacon bytes, and frozen-policy probabilities were recomputed. Runtime nodes check the secret hop tags; the private keys are not exported for public verification. This branch leaves the original 750-run traces, checkpoints and decision rule unchanged.

5.4 Host controlled radio bridge firmware

Four pseudo-terminal devices expose the TX, RX, MAC-confirm and telemetry interface while ns-3 carries the application bytes. Four host-adapter processes use the same serial contract intended for later board integration. The first diagnostic delivered five of ten packets and revealed a timestamp error: the host sampled its clock before a blocking read, then used that earlier time to judge freshness. Sampling again when bytes arrived removed the false rejections. The corrected fixture delivers ten of ten packets and settles service at both relays. Both captures are retained. One functional fixture is insufficient to estimate reliability.

The PTY backend models the firmware interface without executing MCU instructions. Its greedy, MLP and attention rechecks each deliver ten of ten packets. A separate Renode 1.17.0 backend runs compiled C firmware on four emulated CC2538 Cortex-M3 boards (Fig. 7). Firmware parses UART JSON, bounds frame sizes, handles the IEEE 802.15.4 radio FIFO and correlates MAC acknowledgments. Routing inference and cryptography remain on the host. Each control delivers ten of ten packets and returns ten verified source receipts.

Five probes check the uncalibrated telemetry label, exact 114-byte delivery with ACK, rejection of 115-byte frames, malformed-hex rejection and failed out-of-range transmission. A separate capture audit checks MCU UART bytes, public sink signatures and packet identities. Development captures retain the startup and metadata corrections. The stricter audit also rejects an older greedy capture with a missing relay-1 final log, despite its saved summary. New runs pass only after all ten offered identities reach the sink, ten signed receipts are accepted at the source, source and relay final records are complete, all five probes succeed and the capture audit passes. All three fresh controls meet these requirements.

The full attention actor–critic stores 63,368 FP32 parameters, or 253,472 bytes (247.53 KiB). That count includes critic parameters that an actor-only implementation could omit. For comparison, the CC2538SF53 variant provides 512 KB flash and 32 KB RAM [22]. Flash-resident weights need not all occupy SRAM, but flash capacity alone says nothing about activation buffers, stack and protocol memory. An embedded build, linker map and peak-memory measurement are needed before claiming feasibility. The present MCU firmware handles UART parsing, frame bounds, radio FIFO access and MAC acknowledgments; it executes neither neural inference nor HMAC/Ed25519.

Renode's deterministic range medium differs from the ns-3 physical-layer model used for statistical evaluation. Energy telemetry is explicitly labeled emulated_no_power_model: the fraction is fixed, while queue telemetry counts pending MCU ACK records. It provides no battery measurement. The chain offers one feasible next hop at each relay, so it tests integration rather than policy selection or reliability. Deadline debts remain nonzero. The emulation does not validate physical clock and pin initialization, physical RF behavior, embedded policy/crypto execution or calibrated current consumption.

5.5 Post review stress and sensitivity diagnostics

Table 7 and Fig. 9 report a separate exploratory extension on fixtures 84001–84003, using attention checkpoints 41001–41003. Active proofs, censored passive and frozen prior each run for 60 and 300 s, giving 18 runs. Packets arrive every 0.2 s from 0.4 s until near the end of the run: 290 and 1,490 offered packets, respectively. The topology, blackhole fraction, beacon schedule and 0.5 J routing-energy scale follow the frozen benchmark. These diagnostics were designed after review and do not enlarge the original confirmatory test family.

The extension exposes a limit rather than steady operation. At the assumed listening current, 0.5 J corresponds to roughly 8.87 s. Routing excludes an energy-depleted node, but the radio bridge continues telemetry and PHY-state integration. Delivery consequently plateaus while offered traffic and source-service debt continue to grow. Accumulated radio energy after depletion is bookkeeping for an energized simulation, not energy physically supplied by the original battery. Retained-watch counts also persist because closed records are not reclaimed.

Table 7. Long horizon stress diagnostics.

Horizon / arm	Packets	Delivery (%)	Critical failure (%)	Max. source debt
60 s / Active proofs	290	13.45	85.13	3.94
60 s / Censored passive	290	13.10	85.12	3.94
60 s / Frozen prior	290	12.99	85.43	3.94
300 s / Active proofs	1490	2.62	97.22	15.36
300 s / Censored passive	1490	2.55	97.22	15.36
300 s / Frozen prior	1490	2.53	97.29	15.36

Delivery and critical-failure entries are means across three fixtures; debt is the maximum final delay debt across sources and fixtures. Zero handle-exhaustion events were recorded. Pending-map samples are taken once per second and cannot rule out transient MAC queue drops. This small extension supplies no bounded-queue or stability guarantee.

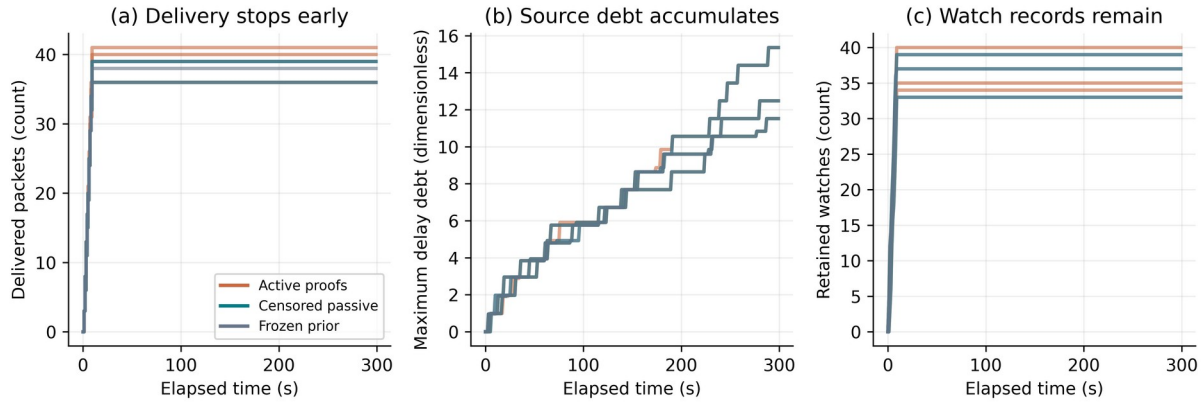


Fig. 9. Executed 300 s stress trajectories.

Figure 10 examines Eq. (1) on three further fixtures, 85001–85003, paired with checkpoints 41001–41003. Each coefficient is changed alone by −20% or +20%; the unchanged setting and ten perturbations give 33 four-second blackhole runs. All 30 perturbed fixture-level delivery contrasts are exactly zero in these runs. This limited result does not establish insensitivity in denser candidate sets or outside the tested range. These are local sensitivity observations, not a tuning procedure or a confidence interval. No checkpoint is retrained and no setting is selected from these results.

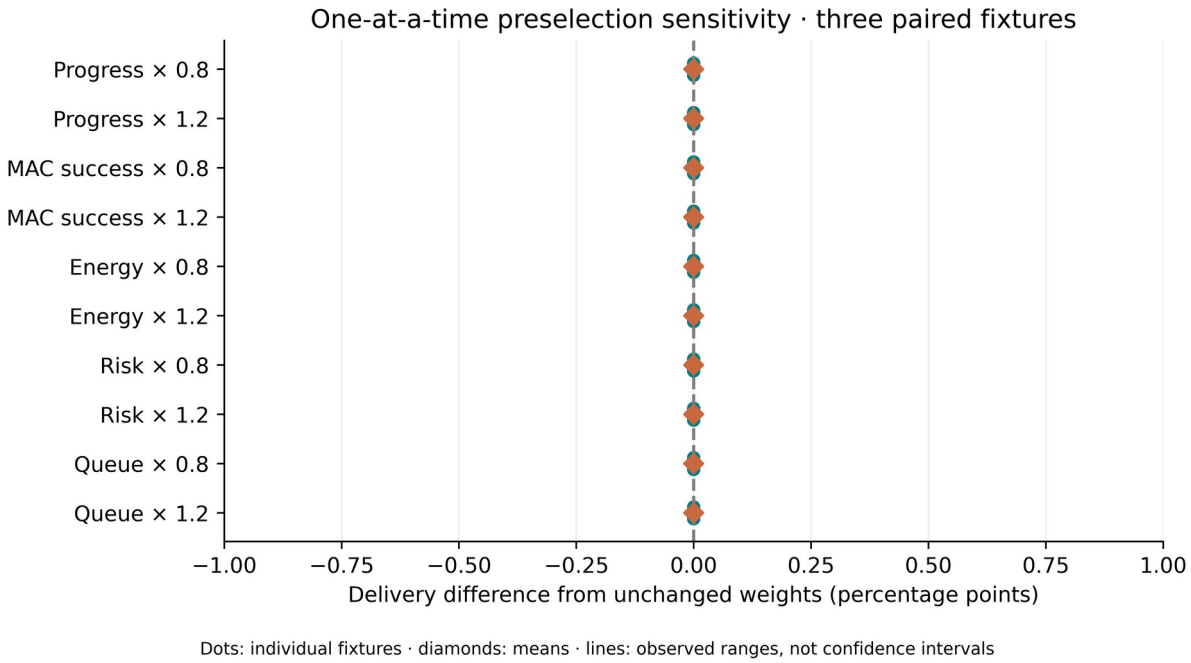


Fig. 10. Executed coefficient sensitivity contrasts.

The full actor–critic contains 63,368 parameters, while the inference actor requires 44,161 parameters in the audited implementation: 176,644 FP32 weight bytes before runtime buffers. This arithmetic corrects treating the complete training network as an unavoidable embedded inference footprint. It does not establish an MCU deployment. Duty-cycled MAC performance, drift tolerance, target inference and measured energy require separate validation.

6. Discussion

6.1 Critical vulnerabilities and limitations

The main result concerns replacement of the tested active-proof mechanism. Feedback eligibility and expiry rules change alongside radio traffic, preventing attribution solely to proof airtime. After correction, neither naive passive nor frozen prior provides evidence for a unique censoring effect. The earlier unsuccessful architecture comparisons remain relevant to that interpretation and are retained in the evidence record.

The frozen benchmark relies on simulator-held geometry, packet history and reverse receipt paths; beacon values are supplied through receive-gated metadata. Source-only service debts leave transit queue control untested. Its threat model also excludes forged receipts, deceptive telemetry, compromised sources, collusion and mobility. Overheard transmission confirms an attempt, while successful delivery at the next hop remains uncertain. The effective-count risk score should be read as a routing preference, not an evaluated live intrusion detector.

A four-second horizon, a central sink and a 60-packet burst are insufficient to establish steady-state scalability or network lifetime. After cache sorting, candidate scoring is linear in the bounded candidate count. Record expiry is more costly: each millisecond poll scans all W retained watches, giving O(W) work and O(W) memory. The experiments provide neither a long-horizon reclamation policy nor measured embedded inference cost. An always-listening radio also omits the wake-up cost that evidence collection would incur under duty cycling.

The 150 ms evidence timeout includes contention, retries and forwarding. Interference can therefore delay a valid receipt past expiry. Censoring avoids assigning failure solely from silence and allows a

later receipt to upgrade the record; it does not establish an optimal timeout. The simulator also uses a shared time base. The corrected host timestamp sampling error is different from device clock offset or drift, which remain untested.

The holdout fixes passive strength at 0.5, record expiry at 150 ms, cache life at 0.75 s, the current model and all inference checkpoints. Sensitivity to those choices remains unknown. Changes to serialization, ACK authentication, service tracking or ledger retirement require a separately frozen experiment; incorporating them after observing the holdout would change the question being tested.

6.2 Future experiments

A useful next step is to decide when forwarding evidence is worth collecting. An observation policy could choose silence, passive listening or an authenticated probe according to the expected reduction in delivery loss relative to measured radio and processing cost. A discriminating experiment would match feedback semantics and data airtime, introduce duty cycling and adversarial receipts, and reserve new topologies and physical-board settings. This would test evidence acquisition itself, which the current comparison cannot isolate.

7. Conclusions

Receipt-assisted passive feedback passed the prespecified replacement test on 30 new simulation fixtures, improving blackhole delivery over active proofs by 4.67 percentage points. The matched controls do not support a unique censoring or attention advantage. A separate 180-run authenticated transfer evaluates frozen policies with received wire state and local service records; it likewise establishes no attention superiority. Serial-interface testing identified and corrected a receive-time error, and four emulated Cortex-M3 boards execute the compiled radio-interface firmware. The 60–300 s extension exposes routing-energy exhaustion and increasing source debt rather than sustained operation. These results support the tested feedback replacement and document how far the implementation has been exercised. Absolute service compliance, physical-board operation, embedded policy/crypto execution and measured energy remain open.

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Data and code availability: code, frozen protocol, checkpoints, historical negative outcomes, raw traces and figure sources are available at https://github.com/DrShivamBhardwaj/JICCE_LA_CPPO_Reproducibility. The pre-outcome source freeze is 4027f071322cb5682f39329f656d76df3b68ff62. The complete earlier evidence manuscript is retained in publication/censored_feedback/.

Declaration of generative AI use

ChatGPT assisted manuscript drafting, code preparation, analysis scripts and programmatic figure plotting. All numerical results reported here come from the retained executable experiments and traces. Responsibility for scientific verification, interpretation and final approval rests with the authors.

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